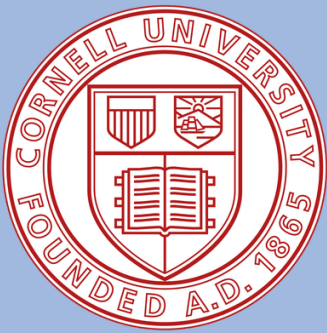




What if your AI could look things up?

Re-implementing Retrieval-Augmented Generation (RAG) · Lewis et al., NeurIPS 2020

DPR retriever + BART seq2seq generator on open-domain QA

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Problem With LLMs

- LLMs struggle on deep knowledge and large memory based tasks; they hallucinate responses because when they don't know the answer, they're trained to priority fluency over accuracy.
- This is why we use a two-step LLM model (architecture to the right) to give LLMs the information they need to answer. The retriever is responsible for retrieving the most relevant, ground-truth documents for the generator (pre-trained seq2seq model) to look through



Limitations on our setup

RETRIEVAL DOC VOLUME

Indexing/Training on 21.5M passages exhausted compute. Used 50k Wikipedia passages instead.

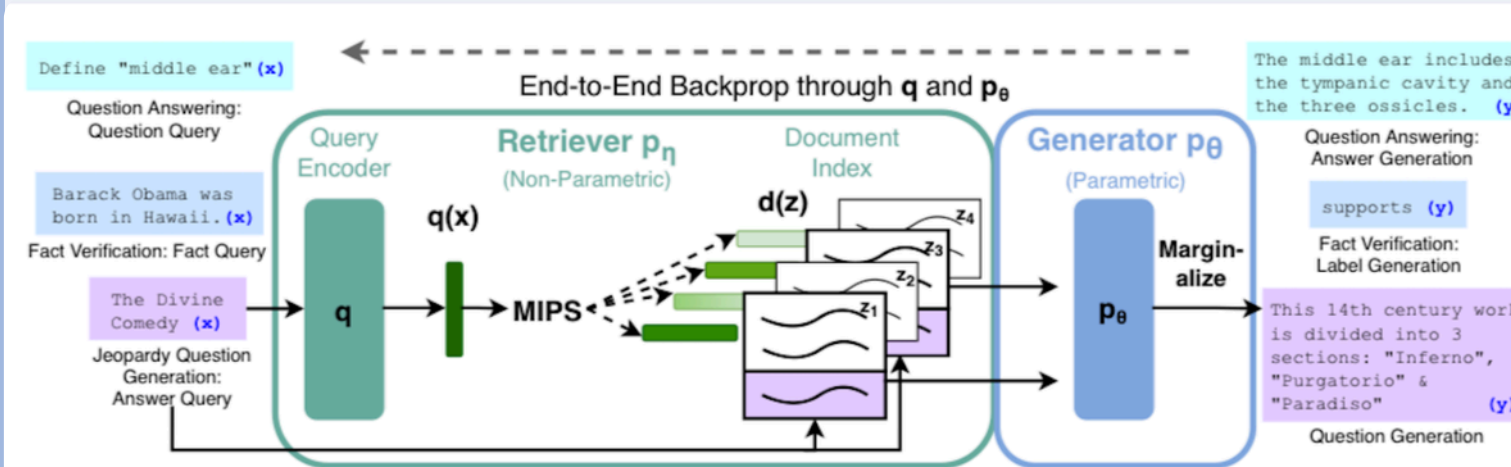
DATASET

For purposes of time and compute, we ran evaluation on 2 of the 4 datasets included in the paper

COMPUTE

While Paper used 64 A100 GPUs, we had access to 1 A100 GPU

RAG Architecture



Methods & Initial Results

- Reproduced **RAG-Sequence with DPR/MIPS retrieval + BART**
- Evaluated with task-specific metrics: **EM for NQ**, and **ROUGE-L/BLEU-1 for MS-MARCO examples**.
 - Filtered out “No Answer Present” examples to better match the abstractive QA setting.
- NQ used 87,925 train / 3,610 dev examples. MS-MARCO used 20k train / 1k dev examples**
 - Dev examples are question-answer examples from development set used for eval
- All experiments trained for 3 epochs

Results Table 1: Baseline values across tasks

Task	Data	Ours	Paper
Open-Domain QA	Natural Questions	11.41% EM	44.5% EM
Abstractive QA	MS-MARCO	34.27 ROUGE-L, 25.26 BLEU-1	40.8 ROUGE-L, 44.2 BLEU-1

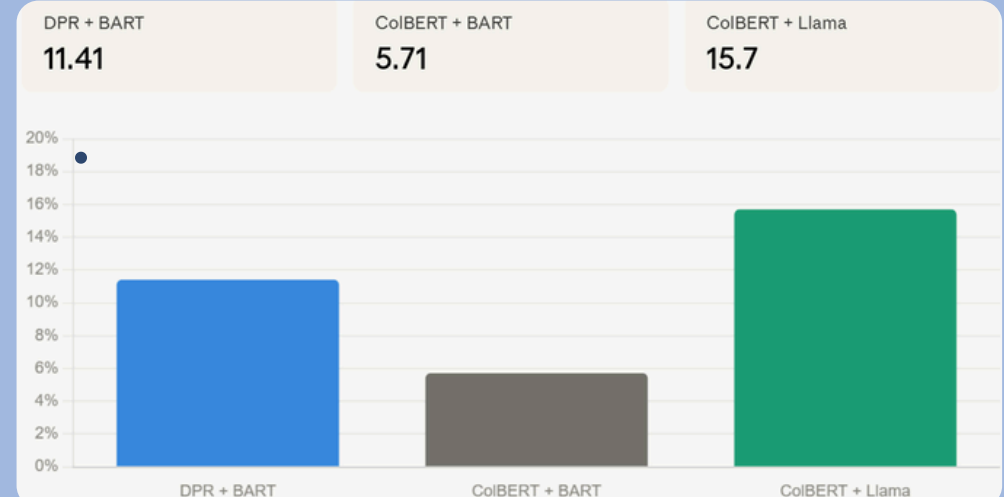
Results Table 2: Qualitative MS-MARCO Output Examples

Example	Prediction	Reference	Takeaway
Good: Can fertilizer burn lawn?	“Yes, you can burn your lawn with fertilizer.”	“Yes, over-fertilizing can burn lawn.”	Correct meaning; concise answer.
Weak: Lightning strikes per day	“The average number of lightning strikes per day is 10.”	“Globally 8,640,000 lightning strikes per day.”	Fluent but factually wrong; struggles with precise numbers.

Qualitative outputs show that RAG-Sequence performs well on simple yes/no-style MS-MARCO questions, but still struggles with precise factual grounding.

Additional Improvements

- Paper uses DPR, good for efficiency, but **ColBERT** provides much more fine-grained token-to-token multi-vector matching,
- The Paper uses an older generator --facebook's BART, while we propose **NVIDIA's Llama-3.1**, built for improved contextual reasoning



Conclusion

- We ultimately find, through our additional improvements, that **RAG performs better on a combination of Colbert + Llama**, even when trained on a tiny fraction of the original data
- An interesting note is the ColBERT + BART result, whose results paled in comparison to the original architecture. This can be explained by us not retraining BART on ColBERT-level architecture, which was necessary for improved results, but wasn't possible due to colab compute limits being exhausted
- Some changes we would make in the future is assessing our compute limits ahead of time to accomodate time limits, as it took us almost a week to decide to train/eval on 50k passages